

Learning under Distribution Shift for Obesity Risk Prediction

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Introduction

Distribution shifts pose significant challenges to the deployment of machine learning models, particularly in high-stakes applications such as autonomous driving and healthcare decision-making. This paper aims to learn a regression model to predict continuous Body Mass Index (BMI), which is robust to distribution shift.

Background: Distributionally Robust Optimization

In supervised learning tasks, we aim to learn a decision rule $h : \mathcal{X} \rightarrow \mathcal{Y}$ on testing distribution Q by minimizing the empirical loss on training distribution P assuming $P = Q$. This would not work when there is distribution shift between P and Q and in reality Q is often unknown. One approach is to apply Distributionally Robust Optimization(DRO) framework. The estimated optimal decision rule $\hat{h}$ can be written as The estimated optimal decision rule $\hat{h}$ can be written as:

$$\hat{h} = \arg \min_{h \in \mathcal{H}} \sup_{Q \in \mathcal{Q}} \mathbb{E}_{(X,Y) \sim Q} [\mathcal{L}(h(X), Y)]. \quad (1)$$

- $\mathcal{H}$: Decision space
- $\mathcal{Q}$: Set of plausible testing distributions
- $\mathcal{L}$: loss function

Specifically, $\mathcal{Q}$ is often defined as: $\mathcal{Q} = \{Q : D(Q||P) \leq \delta\}$, where D represents a distance metric, such as KL-divergence, and δ specifies the allowable deviation.

Background: Rockafellar-Uryasev (RU) Regression

Traditional DRO methods are computationally expensive and do not scale well. Rockafellar-Uryasev (RU) Regression is a new approach within the DRO framework that leverages Conditional Value-at-Risk (CVaR) concepts to achieve robustness [1].

The main assumption is Γ -sampling in the training distribution. In the context of medical risk prediction, Q represents the ground truth distribution within the population, while P denotes the sampling distribution generated from Q via a selection process prone to bias. This paper alternates between bias and distribution shift, because sampling bias is the primary source of distribution shift in our analysis.

Let $S \in \{0, 1\}$ denote a selection indicator. Then, it holds that:

$$\mathbb{P}_Q(Y | X, S = 1) = \mathbb{P}_P(Y | X),$$

and the selection bias is bounded as follows:

$$\frac{\mathbb{P}_Q(S = 1 | X, Y)}{\mathbb{P}_Q(S = 1 | X)} \in [\Gamma^{-1}, \Gamma]$$

for some constant $\Gamma \geq 1$. Γ characterizes the degree of selection bias, where $\Gamma = 1$ signifies no bias and larger values indicate increased flexibility to accommodate greater levels of bias. It can be proved that under this assumption, learning optimal rule $\hat{h}$ in (1) is equivalent to minimizing an augmented convex loss function, called Rockafellar-Uryasev loss[1], which will be defined later.

Dataset

- **Training Data:** Behavioral Risk Factor Surveillance System (BRFSS) 2021–2023 (prone to bias).
- **Ground Truth:** National Health and Nutrition Examination Survey (NHANES) 2021–2023 (physically measured).
- **Data Splits:** Training: 71,448 samples, Validation: 17,863 samples, Testing: 5,169 samples.

Features:

- **Input Features:** Gender, Age, Race, Education, Diabetes, Hypertension, Poverty_Level, Health_Insurance (categorical); Poverty_Index (numerical).
- **Preprocessing:** Categorical features were converted using one-hot encoding. Poverty_Index was calculated based on household income, size, and federal poverty guidelines.

Target Variable:

- **BMI:** Continuous output variable. Values greater than 40 were truncated to 40 to focus on a meaningful range for predictive modeling.

Model architecture

To implement RU regression, I jointly train two neural networks: one learns decision rule h that maps input features to outputs space, while the other learns an auxiliary function α that map from input features to $\mathbb{R}$.

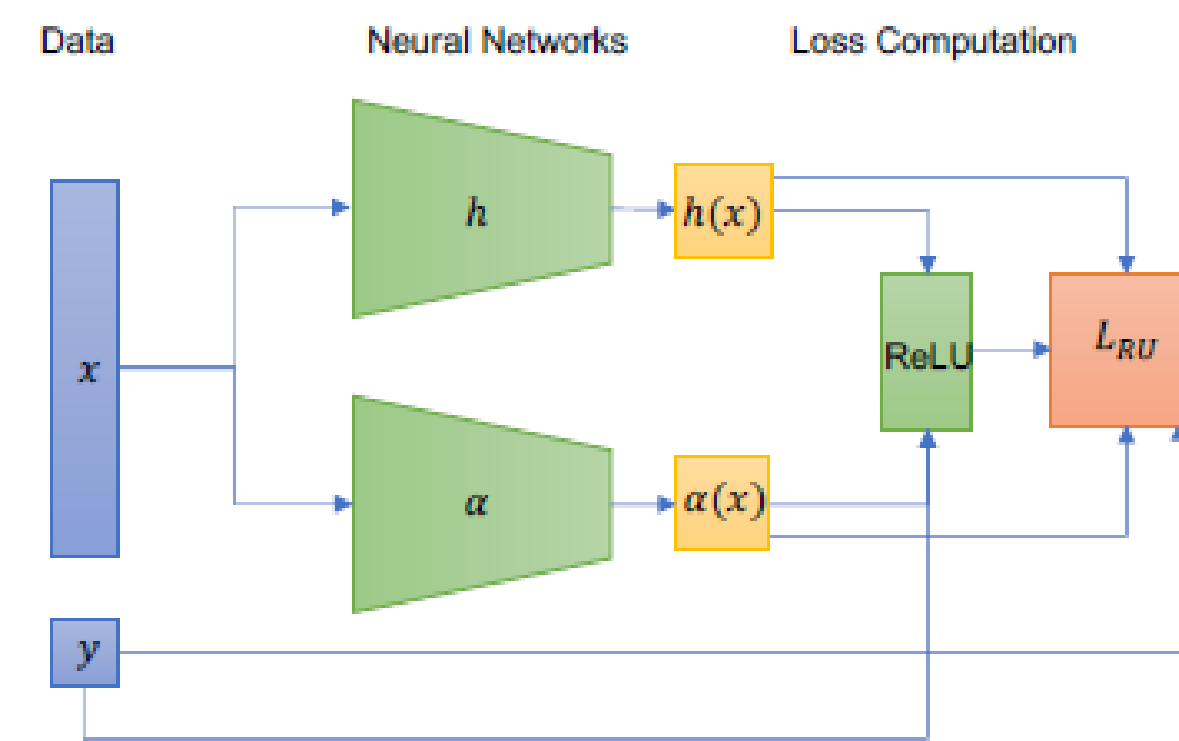


Figure 1. Caption

Baseline Model

For comparison, a neural network with same structure as well as a linear regression model, were trained. Before training, the input data points were reweighted to adjust for covariate shift. Specifically, a logistic regression model was trained on a constructed dataset, where the input features from the training set (X_{train}) were assigned a label of -1 , and the features from the testing set (X_{test}) were assigned a label of $+1$. The logistic regression model learns to distinguish between the two domains. The density ratio $w(x)$ is then estimated as:

$$w(x) = \frac{p(x | \text{test})}{p(x | \text{train})} \approx \frac{\sigma(z(x))}{1 - \sigma(z(x))},$$

where $z(x)$ is the output of the logistic regression model (logit score), and $\sigma(z)$ is the sigmoid function. The weights $w(x)$ are used to reweight the training data points.

Rockafellar-Uryasev loss function

The augmented convex loss function is defined as:

$$\mathcal{L}_{RU}^{\Gamma} = \frac{\mathcal{L}(h(x), y)}{\Gamma} + \left(1 - \frac{1}{\Gamma}\right) \alpha(x) + \left(\Gamma - \frac{1}{\Gamma}\right) \text{ReLU}[\mathcal{L}(h(x), y) - \alpha(x)]$$

- $\mathcal{L}(h(x), y)$: Regular Mean Squared Error (MSE) loss

$$\mathcal{L}(h(x), y) = \frac{1}{n} \sum_{i=1}^n (h(x_i) - y_i)^2$$

- Γ : Bias degree, a hyperparameter to be optimized.
- $\alpha(x)$: auxiliary function learned alongside the decision rule h.This helps estimate the worst-case loss under potential distribution shifts.

Result/Future work

Root Mean Squared Error (RMSE) was used as the evaluation metric on the test set, as summarized in Table 1.

Method	Train RMSE	Test RMSE
NN Model	5.27	6.61
Linear Model	5.26	6.61
NN Model(Adjusted)	5.26	6.53
Linear Model(Adjusted)	5.26	6.38
RU ($\Gamma = 1.5$)	5.60	6.17
RU ($\Gamma = 2$)	5.60	6.09
RU ($\Gamma = 3$)	5.60	6.10
RU ($\Gamma = 4$)	5.73	6.03
RU ($\Gamma = 5$)	5.63	6.05
RU ($\Gamma = 10$)	5.86	6.05

Table 1. Comparison of model performance using RMSE

RU regression models underperform during training but outperform during testing compared to baseline model with and without covariate shift correction, indicating robustness to distribution shifts. The model with $\Gamma = 4$ achieved the best test performance despite poorer training results, likely due to stronger bias correction with higher Γ . However, as Γ continues to increase, perform starts to decline due to excessive bias correction and thus reduced fitting on the data.

Future work: Predicting individual BMI from the available data is inherently challenging, as evidenced by the large errors in all models. Subgroup analyses at the state level could offer valuable insights for localized policy-making. However, NHANES lacks state-level data. Future research could explore alternative datasets with subgroup information to enable more granular analyses. Another direction of exploration could focus on high-dimensional data, such as those from medical imaging or language tasks, to further evaluate RU regression’s capacity to handle distribution shifts in more complex scenarios.

References

- [1] Roshni Sahoo, Lihua Lei, and Stefan Wager. Learning from a biased sample. *arXiv preprint arXiv:2209.01754*, 2022.